

DEVA ANAND

devaanand@umass.edu • (413) 315-1715 • devaanand.com • linkedin.com/in/devaaa • github.com/Deva-1903

Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027
Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** via Mayden iaptus REST integrations; contributed to **\$340K+** in revenue.
- Architected a full-stack internal tooling platform (React, Node.js, MongoDB) used daily by the AI team, cutting manual work by **100+ hours/month** and lifting throughput by **60%**.
- Hardened backend security and reliability: **AWS KMS** encryption for sensitive data, auth and rate-limit hardening, and **20+ VAPT** remediations, with integration tests for validation and production log monitoring for incident visibility.
- Served as primary technical POC, wrote knowledge-transfer documentation onboarding a second developer in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot experiment for AI-assisted development.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM feature (**4-bit quantized Llama-7B via llama.cpp in C/C++**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with **git-webhook build/deploy automation** (GitHub webhook to a droplet on push), cron automation, and Stripe payments; live **3+ years**.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI backend service with automated provider routing/fallback and pipeline instrumentation for **log visibility**; validated and hardened with **150+ automated tests**.

AutoEval – Retrieval Evaluation Improvement Agent (GitHub) Python, IR evaluation, Git automation

- Built an automation tool that proposes and validates changes, scores them against multiple reference systems, and commits only improvements to **Git** with report cards and guardrails, keeping validation reproducible and auditable.

Cache- and SIMD-Aware Matrix Multiplication (GitHub) C++, AVX/SIMD, OpenMP, Linux perf

- Optimized C++ matrix multiplication (systems programming) through cache-aware tiling, **AVX/SIMD** vectorization, and **OpenMP**, measuring and validating L1/L3/TLB performance with **perf**.

Publications

- “Alzheimer’s Disease Classification using Transfer Learning,” IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, C++, TypeScript, JavaScript, SQL, Java, Bash, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: FastAPI, Node.js, Express, React, Docker, Git, GitHub Actions, REST APIs

Automation / Testing: pytest, integration testing, CI/CD, Git automation, git-webhook deploy, perf profiling, debugging

Backend / Observability: REST APIs, authentication, rate limiting, observability (logging / monitoring), Linux, security

Cloud / Platforms: AWS, GCP, Azure, Docker, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.